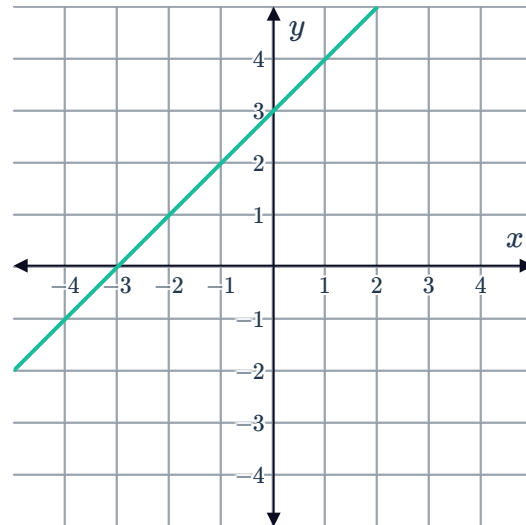


Review: Equations of lines



- 1 Explain why the following statements are incorrect.
 - a "A line with an undefined slope has no equation."
 - b "The slope of the line with equation $7x + y = 3$ is 7."
- 2 A line has the equation $y = 6(3x - 2)$.
 - a Rewrite $y = 6(3x - 2)$ in the form $y = mx + c$.
 - b State the slope and y -intercept of the equation.
- 3 Write the equation of the following lines in slope-intercept form:
 - a A line whose slope is -8 and crosses the y -axis at -9 .
 - b A line that has a slope equal to -7 , and cuts the y -axis at 7 .
 - c A line that passes through the point $A(-2, -9)$ and has a slope of -2 .
- 4 The following is a graph of $y = mx + b$.
Sketch the graph of y when b is decreased by 5.



- 5 Line L_1 has the following equation: $y = x - 2$
 - a Find the y -value of the y -intercept of the line.
 - b Find the x -value of the x -intercept of the line.
 - c Find the y -coordinate of a point that has an x -coordinate of -5 .
 - d Sketch the line $y = x - 2$ on the coordinate plane.

6 Express the following equations in standard form.



a $y - 5x = 5$

b $y = 2x - 5$

c $y = \frac{2x}{5} + 4$

7 A line has slope $\frac{14}{3}$ and passes through the point $(-5, -10)$.

a Find the value of b for this line.

b Write the equation of the line in standard form.

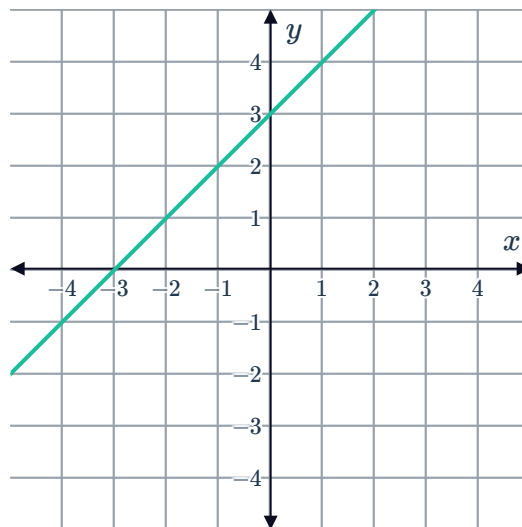
8 Determine whether the following could be the equation of the graphed line.

a $2x - 2y = 6$

b $2x - 2y = -6$

c $2x + 2y = 6$

d $2x + 2y = -6$



9 $A(-1, 4)$ is a point on a straight line. If the slope of the line is 4, determine whether the following statements are correct.

a $\frac{x+1}{y-4} = 4$

b $\frac{y-4}{x+1} = 4$

c $\frac{y+1}{x-4} = 4$

d $\frac{4-y}{-1-x} = 4$

10 A line passes through the points $(3, -5)$ and $(-7, 2)$.

a Find the slope of the line.

b Find the equation of the line in slope-intercept or standard form.

11 Use the two point formula to derive the equation of the line that passes through the points $A(8, 1)$ and $B\left(-2, \frac{1}{7}\right)$. Give your answer in standard form.

- 12 A line has an x -intercept of 8 and a y -intercept of 5.



By considering the intercept form of a straight line, write the equation of the line in the standard form.

13

- a Find the equation, in standard form, of the line that passes through $A(7, -14)$ and $B(-7, -15)$.
- b Find the x -coordinate of the point of intersection of the line that goes through A and B , and the line $y = x - 8$.
- c Hence, find the y -coordinate of the point of intersection.

Additional questions

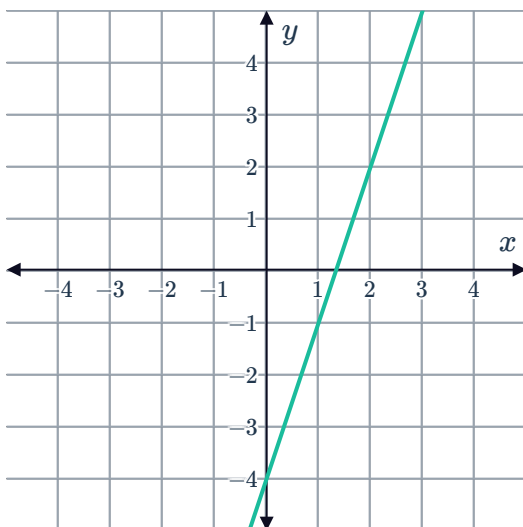
- 14 For each of the following linear equations:

- i State the value of m , the slope.
 - ii State the value of b , the y -intercept.
- a $y = 6x + 4$
 - b $y = -3x - 8$

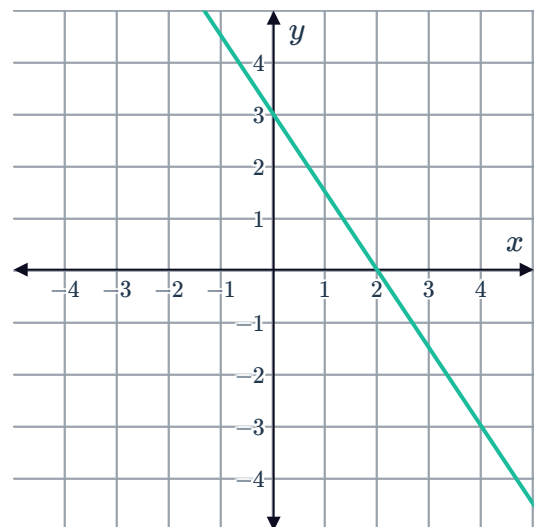
- 15 For each of the following graphs:

- i Determine the slope of the line.
- ii Determine the y -intercept of the line.
- iii Determine the equation of the line.

a



b



16 Consider the following linear equations:



- Line 1: $y = 2 + 6x$
- Line 2: $y = 5x + 6$
- Line 3: $y = 6x$

- Line 4: $y = \frac{x}{6} + 2$
- Line 5: $y = 5 - 6x$
- Line 6: $y = 6x - 2$

State the three lines that have the same slope.

17 For each of the following lines, state whether the following lines have a y -intercept or not. If the line does have a y -intercept, state the coordinates of it.

- a $y = 2$ b $5x = 4y$ c $y = -4x + 1$ d $x = 1$

18 Suppose the Line L_1 has the equation $y = \frac{-9x + 3}{3}$.

- a Simplify L_1 to the form $y = mx + b$.
- b State the slope of the line.
- c Find the y -value of the y -intercept of the line.

19 Determine whether each of the following lines has the same slope as $y = 6x + 3$ or not.

- a $y = 6x$ b $y = 3x + 3$ c $12x - 2y = 3$ d $18x + 3y = 0$

20 Find the equation of the lines matching the following descriptions:

- a A line that has the same slope as $y = 2 - 4x$ and the same y -intercept as $y = -7x - 3$.
- b A line whose slope is $\frac{2}{3}$ and goes through the point $(0, 5)$.
- c A line that is horizontal and goes through the point $(9, -2)$.
- d A line that is vertical and goes through the point $(11, -9)$.

21 Sketch the following lines on a coordinate plane:

- a The line that has a y -intercept of -4 and whose slope is $-\frac{4}{3}$
- b The line $y = 2x + 4$
- c The line $y = -2x + 4$
- d The line $y = \frac{1}{2}x - 1$
- e The line $x = -3$

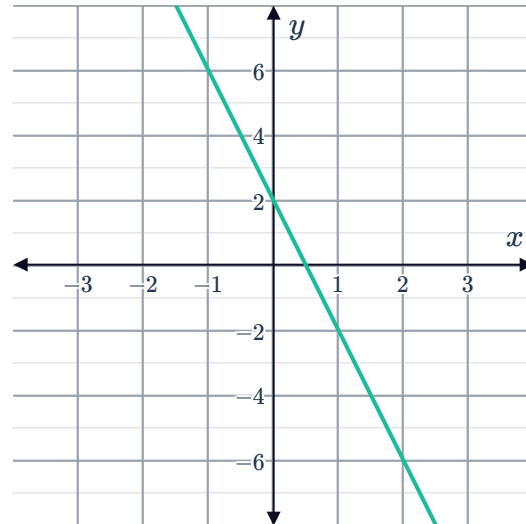
22 Consider the following line:



- a** Copy and complete the following table of values.

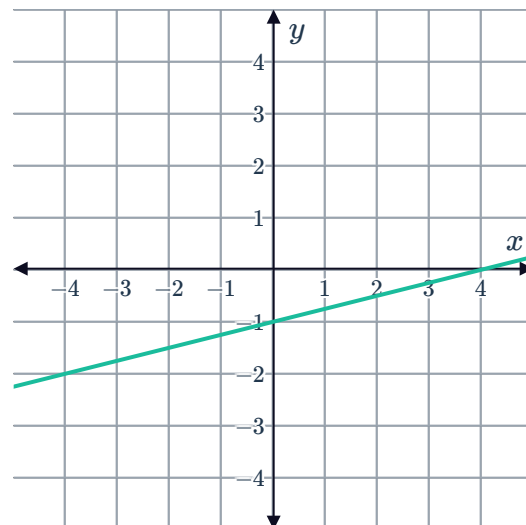
x	-1	0	1	2
y				

- b** State the values of the slope, m and the y -intercept b .
- c** Write the linear equation expressing the relationship between x and y .



23 Consider the following line:

- a** State the values of the slope, m and the y -intercept b .
- b** Write the linear equation expressing the relationship between x and y .



24 For the following lines passing through the given two points:

- i** Find the slope of the line.
- ii** Find the equation of the line in the form $y = mx + b$.

a $(0, 5)$ and $(2, 9)$

b $(-3, 9)$ and $(4, 2)$

c $(-6, 8)$ and $(6, 0)$

d $(-7, 3)$ and $(-7, -2)$

25 State the standard form of a linear equation.

26 State which of the following linear equations are in standard form:

a $2x - 5y = 0$

b $x = 10$

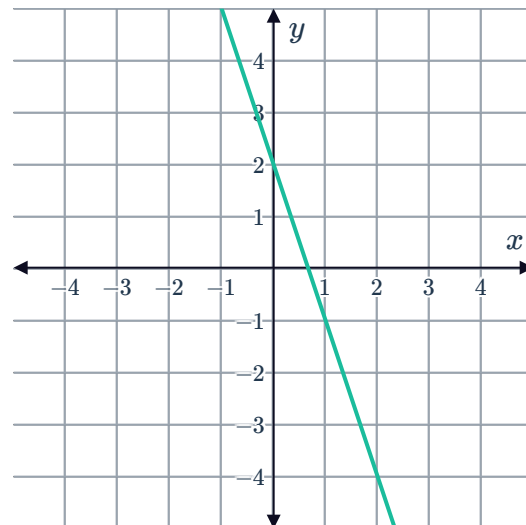
c $x = \frac{1}{m} + 5$

d $2x + 7y = 11$



27 Consider the following line:

- a** State the y -intercept.
- b** Find the slope, m , of the line.
- c** Find the equation of the line in the form $y = mx + b$.
- d** Rewrite the equation of the line in standard form.



28 State the point-slope form of a linear equation.

29 For each of the following equations, state the form they are written in:

- a** $y = -4x + 5$
- b** $y - 1 = 5(x - 4)$
- c** $5x - 3y = 8$

30 Write the equation of the following lines in point-slope form:

- a** A line passing through the point $(7, 1)$ has a slope of 4.
- b** A line passing through the point $(4, 0)$ has a slope of -5 .
- c** A line passing through the point $(1, 6)$ has a slope of 0.

31 For each of the following lines:

- i** Find the slope, m , of the line.
 - ii** Write the equation of the line in point-slope form.
-
- a** A line passes through the two points $(7, 6)$ and $(9, 12)$.
 - b** A line passes through the two points $(4, -8)$ and $(2, 0)$.

32 A line has slope $-\frac{3}{2}$ and passes through the point $(2, -2)$.



- a Find the value of b for this line.
- b Hence, write the equation of the line in slope-intercept form.
- c Derive the same equation by using the point-slope formula.
- d Sketch the graph of the line.